

DEXTER FERNANDES

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SUMMARY

Machine learning engineer with 3+ years delivering computer vision and multimodal ML systems into 24/7 production, from data ingestion and training through model serving, monitoring and drift tracking. Led a four-engineer team building real-time multi-camera perception, and owned low-latency streaming inference pipelines in Python and C++17 on NVIDIA DeepStream, TensorRT and OpenVINO. Comfortable across MLOps tooling, experimental design and cost, latency and scale trade-offs on cloud and edge targets, with recent project work on retrieval-augmented generation and LLM evaluation. MSc Robotics (Merit), University of Bristol.

RELEVANT EXPERIENCE

Esbaar – AI Engineer

01/2023 – 08/2023 · MUSCAT, OMAN

- Owned the end-to-end machine learning lifecycle for real-time perception on BP drilling rigs, from data curation and training through deployment and on-site validation, improving F1 by 15% under continuous 24/7 operation.
- Built ingestion and validation pipelines across image, video and sensor data with Apache Airflow, MLflow, DVC and CVAT, processing 3.6 million records and cutting rollout from 72 hours to under 24 hours.
- Developed NVIDIA DeepStream 6.x and FFmpeg pipelines for continuous streaming inference, hardened with Dockerised microservices, Kafka messaging and health checks, cutting live video pipeline crashes by 67%.
- Owned post-deployment monitoring and drift tracking for fielded vision systems, catching regressions early and feeding failure analysis back into retraining.
- Adapted YOLOv5 detectors in PyTorch with Optuna sweeps and Weights & Biases experiment tracking, and added pytest, unittest and linting to CI for edge inference modules.
- Built an activity classification workstream over 91 sensor channels using MultiRocket, gradient boosting and random forests for multivariate time-series classification.

LivNSense Technologies – Lead Software Engineer, AI

05/2020 – 12/2022 · BENGALURU, INDIA

- Led four engineers building a two-stage detector feeding helmet-colour classification for construction safety in Texas, cutting false positives by 33% within latency and power budgets.
- Set experimentation standards, reviewed model performance and mentored junior engineers, guiding optimisation strategy for production deployments.
- Built a real-time multi-camera perception pipeline on a moving construction vehicle, synchronising streams and adding ByteTrack multi-object tracking to hold worker and vehicle identities across frames and cut duplicate incident alerts.
- Built asynchronous C++17 inference pipelines on OpenVINO across Intel Movidius VPUs and Core i7 industrial edge boxes, overlapping preprocessing, inference and post-processing to raise throughput from 5 to 15 FPS.
- Optimised architectures with quantisation-aware training, structured pruning and layer fusion in Keras and OpenVINO, cutting model size by 50% while holding production accuracy.
- Deployed containerised ML inference microservices on Microsoft Azure with REST endpoints and queue-based async processing, owning autoscaling, monitoring and cost-per-inference.

PROJECTS

DeepStream RTSP Pipeline github.com/Dexter-Fernandes/deepstream-rtsp-pipeline

06/2026

- Developed an end-to-end DeepStream 9.0 and GStreamer RTSP pipeline running YOLO detection and NvDCF tracking across 3 concurrent live feeds, with metadata extraction, anonymisation and re-streamed output.
- Converted YOLO26n to a dynamic-batch TensorRT FP16 engine, validated FP16 against FP32 accuracy and profiled p99 latency to sustain 29.7 FPS per stream live.
- Benchmarked IoU, NvDCF and NvSORT trackers on MOTA, IDF1, ID switches, fragmentation, FPS and VRAM impact to guide tracker selection.

FinanceBench RAG Evaluation Harness

06/2026

- Developed and evaluated a retrieval-augmented generation harness for document question answering, separating retrieval, generation and grounding failures rather than reporting a single opaque accuracy figure.
- Built a hybrid retrieval pipeline combining BM25, FAISS dense embeddings and reciprocal rank fusion with cross-encoder reranking, using three isolable retrieval modes to separate retrieval quality from generation quality.
- Fine-tuned a bge-reranker-base cross-encoder with LoRA and hard-negative mining, self-hosted behind a containerised inference API, and benchmarked it against the pretrained baseline on MRR, nDCG, evidence hit rate and downstream answer accuracy.
- Built a stateful LangGraph agent orchestrating retrieval, correctness and grounding tools, and paired deterministic checks with LLM-as-judge scoring, benchmarking diagnostic accuracy, latency and tool usage against a deterministic baseline.

Modern Edge CV Benchmarking github.com/Dexter-Fernandes/ModernEdge-CV-Benchmarking

03/2026

- Benchmarked modern PyTorch detectors including YOLOv8, YOLOv10, YOLO26 and RT-DETR on accuracy, latency and model size to compare deployment suitability.
- Evaluated Vision Transformer and foundation-model approaches across detection, segmentation, depth estimation and open-vocabulary understanding, exporting to ONNX to test deployment paths.

EDUCATION

University of Bristol – MSc Robotics, Merit

Dissertation: Radiological SLAM with LiDAR Odometry (academic research project)

■ Built a real-time multi-sensor perception system in C++17 and ROS2 on a Clearpath Husky, fusing Velodyne VLP-16 LiDAR, RealSense D435i depth and radiological sensing for inspection mapping.

■ Ran the full perception and mapping stack on an NVIDIA Jetson AGX Xavier, working within the platform's compute and memory limits for on-robot operation.

■ Benchmarked RTAB-Map, LIO-SAM and Cartographer across matched rosbag2 sequences under a single evaluation protocol to compare mapping and localisation behaviour.

09/2023 – 09/2024 · BRISTOL, ENGLAND

Visvesvaraya National Institute of Technology – BTech Electrical and Electronics Engineering

Foundation in signals, systems, embedded hardware and control, which later supported work in real-time machine learning and sensor systems.

07/2016 – 05/2020 · NAGPUR, INDIA

SKILLS

Languages	Python, C++17, SQL, Bash, CUDA C++
Machine learning	PyTorch, TensorFlow, Keras, scikit-learn, HuggingFace, convolutional neural networks (CNNs), Vision Transformers, MultiRocket, gradient boosting, random forests, multivariate time-series classification, Pandas, NumPy
Computer vision	Object detection, object tracking, semantic and instance segmentation, key-event and incident detection, multi-camera synchronisation, optical flow, anomaly detection, OpenCV, YOLO (v5/v8/v10/v26), RT-DETR
Generative AI	Retrieval-augmented generation (RAG), LLM-as-judge evaluation, LoRA fine-tuning, cross-encoder reranking, FAISS, BM25, hybrid retrieval, LangGraph agents, foundation models, open-vocabulary understanding
Streaming inference	NVIDIA DeepStream 6.x/9.0, GStreamer, FFmpeg, RTSP, multi-stream video analytics, metadata extraction, ByteTrack, NvDCF, NvSORT, low-latency real-time pipelines
Model serving	TensorRT, OpenVINO, ONNX, dynamic batching, quantisation-aware training, FP16/INT8, structured pruning, layer fusion, latency and throughput profiling, NVIDIA Jetson AGX Xavier, Intel Movidius VPU
MLOps	MLflow, DVC (Data Version Control), Apache Airflow, Apache Kafka, Weights & Biases, Optuna, CVAT, CI/CD for ML, drift tracking, production model monitoring, benchmarking harnesses, ablation studies, hypothesis testing, Bayesian optimisation
Cloud and production	Microsoft Azure (Container Apps, Blob Storage, queues), Docker, Git, pytest, gtest, FastAPI, REST API design, containerised microservices, autoscaling, cost-per-inference, PostgreSQL, structured logging, observability, Linux
Leadership	Technical leadership, mentoring, experimental design, Agile and Scrum, Jira, stakeholder communication, cross-functional delivery

ADDITIONAL EXPERIENCE

GrowthStage – Technical Assessor, Computer Vision and Software Engineering

Assesses candidate technical depth across perception, machine learning and deployment for engineering hiring.

05/2026 – PRESENT · REMOTE

Taco Bell – Service Champion / Team Trainer

Onboards new staff and handles escalations in a customer-facing role after progressing to Team Trainer.

09/2024 – PRESENT · BRISTOL, ENGLAND

CERTIFICATIONS AND AWARDS

Deep Learning Specialisation, deeplearning.ai (07/2019)